The W1 algorithm: dual-operator polishing with rank fusion for reachability ranking

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1 Problem and notation

Let $G = (V, E)$ be an undirected graph with $n = |V|$ vertices and $m = |E|$ edges, in which every edge $(i, j) \in E$ is present independently with probability $p_{ij} \in (0, 1]$, and let $x \in \mathbb{R}^n_{>0}$ with $\sum_i x_i = 1$ be a prior distribution over the vertices. In a sampled world $G' \sim G$, let $C(i)$ denote the connected component of vertex i. The task is to rank the vertices by their *expected reachable mass*

$$y^*_i = \mathbb{E}_{G' \sim G}\Big[\sum_{v \in C(i)} x_v\Big],$$

which, for uniform x, equals the expected spread $\sigma(\{i\})/n$ of a single seed in the independent cascade model on G. Computing y^* exactly is #P-hard.

We use the following notation. We write $j \sim i$ when $(i, j) \in E$. For a vector v with positive sum, $\mathcal{N}(v) = v/\sum_i v_i$. The function $\mathrm{clip}(u, \ell, h) = \min\{h, \max\{\ell, u\}\}$ restricts u to the interval $[\ell, h]$. The vector $r(v) \in \mathbb{R}^n$ holds the ranks of v in ascending order (the smallest value has rank 1 and the largest rank n); before ranking, the values are rounded to integer multiples of $10^{-12} \max_j |v_j|$, and values that coincide after rounding are ties and receive their average rank. Operations between vectors, such as a_j/x_j, are taken componentwise.

2 The algorithm

Algorithm 1 combines two propagations. Phase A runs a normalised damped diffusion on the escape-scaled pull operator EW to its fixed point, with a scalar damping factor given by the calibrated rule of the earlier study. Its result is embedded in the excess scale and becomes the starting point of Phase B, which performs a few steps of raw propagation with a constant damping factor on the operator EP. The output is the weighted average of the ranks of the two phases.

Algorithm 1 W1 — dual-operator polishing with rank fusion

Require: graph $G=(V,E)$; edge probabilities $p_{ij}\in(0,1]$; prior $x>0$ with $\sum_i x_i=1$
Ensure: score $s\in\mathbb{R}^n$; vertices are ranked by decreasing s
Parameters: c_0, c_1, $[\alpha_{\min},\alpha_{\max}]$, ε, $T_{\max}$, γ, α_B, K, β (Table 1)

Operators and escape probability
for all $(i,j)\in E$ **do**
  $w_{ij}\leftarrow w_{ji}\leftarrow -\ln(1-p_{ij})$ ▷ edge weight
for all $i\in V$ **do**
  $d_i\leftarrow\sum_{j\sim i} w_{ij}$
  $P_{\mathrm{esc},i}\leftarrow 1-e^{-d_i}$ ▷ $=1-\prod_{j\sim i}(1-p_{ij})$
$W_{ij}\leftarrow w_{ij}/d_i$ for $j\sim i$, otherwise 0 ▷ row-stochastic
$EW\leftarrow \mathrm{diag}(P_{\mathrm{esc}})\,W,\quad EP\leftarrow \mathrm{diag}(P_{\mathrm{esc}})\,P$ ▷ $P_{ij}=p_{ij}$
Phase A: diffusion with the calibrated rule to the fixed point
$\bar{P}_{\mathrm{esc}}\leftarrow\frac{1}{n}\sum_i P_{\mathrm{esc},i}$
$\alpha_A\leftarrow \mathrm{clip}\left(c_0+c_1\bar{P}_{\mathrm{esc}},\ \alpha_{\min},\ \alpha_{\max}\right)$ ▷ one number for all of G
$y\leftarrow x;\quad t\leftarrow 0$
repeat
  $y'\leftarrow\frac{1}{2}\left[y+(1-\alpha_A)\,x+\alpha_A\,EW\,y\right]$ ▷ lazy step
  $y'\leftarrow\mathcal{N}(y')$
  $\delta\leftarrow\|y'-y\|_1;\quad y\leftarrow y';\quad t\leftarrow t+1$
until $\delta<\varepsilon$ **or** $t=T_{\max}$
$a\leftarrow y$
Embedding in the excess scale
$z\leftarrow a\Big/\left(\frac{1}{n}\sum_j a_j/x_j\right)$ ▷ $\frac{1}{n}\sum_j z_j/x_j=1$
$b\leftarrow x+\gamma z$ ▷ $\frac{1}{n}\sum_j (b_j-x_j)/x_j=\gamma$
Phase B: K steps of raw propagation
for $k=1,\dots,K$ **do**
  $b\leftarrow(1-\alpha_B)\,x+\alpha_B\,EP\,b$ ▷ no normalisation
Rank fusion
$s\leftarrow(1-\beta)\,r(a)+\beta\,r(b)$
return s

Table 1: Parameters of W1.

Symbol	Value	Role
$c_0,\ c_1$	0.502, 0.558	coefficients of the calibrated damping rule of Phase A
$[\alpha_{\min},\alpha_{\max}]$	$[0.4,1]$	admissible interval of α_A
ε	10^{-12}	convergence tolerance of Phase A (L_1 norm)
$T_{\max}$	20000	maximum number of Phase-A iterations
γ	16	mean excess ratio of the Phase-B starting point
α_B	0.7	constant damping factor of Phase B
K	3	number of Phase-B steps
β	0.7	weight of Phase B in the rank fusion

The parameters γ,α_B,K,β and the Phase-B operator were selected by a staged search on the development panel (180 graphs with $n=100$) and recorded in the file `results/mixing/selected_mixing.json`, together with the file's SHA-256 digest, before any evaluation on the confirmation panels. The values $c_0,c_1,\alpha_{\min},\alpha_{\max}$ are those of the calibrated rule of the earlier study and were not refitted.

3 Remarks

1. **Common origin of the weights and the escape probability.** Since $\ln\prod_{j\sim i}(1-p_{ij})=-\sum_{j\sim i}w_{ij}=-d_i$, we have $P_{\mathrm{esc},i}=1-e^{-d_i}$: the probability that at least one edge of i is present is a function of its weighted

degree in W. For an isolated vertex ($d_i = 0$), row i of W, EW and EP is zero.

2. **Lazy step.** Since $\sum_i y_i = 1$, we have $(1-\alpha_A)x = (1-\alpha_A)x\mathbf{1}^\top y$, so the fixed point of Phase A is the Perron vector of $B = (1-\alpha_A)\,x\mathbf{1}^\top + \alpha_A\,EW$. For $\alpha_A < 1$ the matrix B is positive, so its Perron root is simple and strictly dominant. On bipartite graphs, however, B has an eigenvalue close to $-\rho(B)$ when α_A is close to 1, and equal to $-\rho(B)$ when $\alpha_A = 1$, so that the plain power method converges slowly or oscillates. The iteration with the matrix $\frac{1}{2}(I+B)$ has the same fixed point and converges in these cases as well.

3. **Scales of the two phases.** Phase A is normalised at every step, so only the direction of a matters; line 17 brings it back to the scale of x. Phase B is affine and is not normalised. After K steps

$$b^{(K)} = (\alpha_B\,EP)^K\,b^{(0)} + (1-\alpha_B)\sum_{k=0}^{K-1}(\alpha_B\,EP)^k\,x,$$

so multiplying the starting point $b^{(0)}$ by a constant changes the relative weight of the two terms and hence the ranking. For this reason the starting point $b^{(0)} = x + \gamma z$ is not normalised (for example, to $(x+\gamma z)/(1+\gamma)$): its scale is part of the parametrisation and is fixed by the convention of the excess scale, namely that the mean ratio $(b_j^{(0)} - x_j)/x_j$ equals γ.

4. **Numerical implementation.** For $K = 3$ the raw recursion is computed directly in double-precision arithmetic. For larger K the implementation represents b as a unit vector and a logarithmic scale factor, which avoids overflow without altering the affine recursion. Inside the logarithms of line 2, p_{ij} is capped at $1 - 10^{-12}$, so that an edge with $p_{ij} = 1$ has a finite weight; EP uses the unmodified probabilities. Before line 17, every a_j is replaced by $\max\{a_j, 10^{-300}\}$.

5. **Output.** The score s is not an estimate of y^*; it serves only to order the vertices. The fusion is performed on ranks rather than values, because a and b live on different scales.

6. **Boundary case.** When $\bar{P}_{\text{esc}} \geq (1-c_0)/c_1 \approx 0.892$, the rule gives $\alpha_A = \alpha_{\max} = 1$ and the restart term $(1-\alpha_A)x$ vanishes. Phase A then computes the Perron vector of the lazy EW, which in general does not depend on x; on a disconnected graph it concentrates on the component with the largest spectral radius, and the Phase-A scores of the other vertices decay towards zero, so that they may tie in $r(a)$. On lattice graphs convergence may not be reached within $T_{\max}$ iterations.

4 Complexity

Building the operators takes time $O(n+m)$. Every Phase-A iteration and every Phase-B step is one sparse matrix–vector product of cost $O(n+m)$, and computing the ranks takes $O(n\log n)$. If t_A is the number of Phase-A iterations, the total time is

$$O\big((t_A + K)(n+m) + n\log n\big),$$

and the space is $O(n+m)$. The algorithm is deterministic and requires neither training nor the sampling of worlds.